

11 (short) Two straight infinitely long wires are placed in the same plane and are parallel to each other as shown in Fig. 9. The distance between the two wires is $L = 40$ cm. The two wires each carry a current $I = I_1 = I_2 = 20$ A but with opposite directions of flow.

(a) What is the direction and magnitude of the magnetic flux density at a point midway between the two parallel wires (in the same plane)? [4]

(b) What is the magnetic flux passing through a rectangle in the plane of the two wires that is placed in the middle with $L_1 = L_2 = 10$ cm, and $d = 25$ cm? [6]

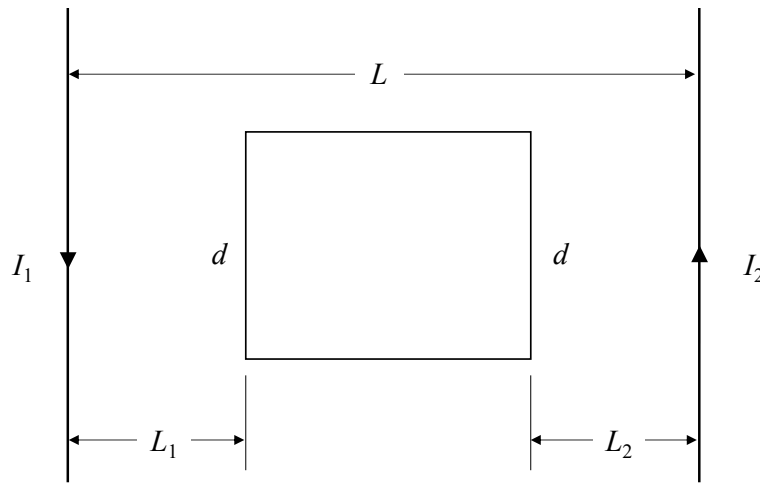


Fig. 9